

Incorporation of Pseudouridine Into mRNA Yields Superior Nonimmunogenic Vector With Increased Translational Capacity and Biological Stability

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In vitro-transcribed mRNAs encoding physiologically important proteins have considerable potential for therapeutic applications. However, in its present form, mRNA is unfeasible for clinical use because of its labile and immunogenic nature. Here, we investigated whether incorporation of naturally modified nucleotides into transcripts would confer enhanced biological properties to mRNA. We found that mRNAs containing pseudouridines have a higher translational capacity than unmodified mRNAs when tested in mammalian cells and lysates or administered intravenously into mice at 0.015–0.15 mg/kg doses. The delivered mRNA and the encoded protein could be detected in the spleen at 1, 4, and 24 hours after the injection, where both products were at significantly higher levels when pseudouridine-containing mRNA was administered. Even at higher doses, only the unmodified mRNA was immunogenic, inducing high serum levels of interferon- α (IFN- α). These findings indicate that nucleoside modification is an effective approach to enhance stability and translational capacity of mRNA while diminishing its immunogenicity *in vivo*. Improved properties conferred by pseudouridine make such mRNA a promising tool for both gene replacement and vaccination.

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INTRODUCTION

In vitro-synthesized mRNA seems an ideal nonviral gene replacement tool with many inherent advantages, including rapid protein production and efficient transduction of primary cells (reviewed in ref. 1). mRNA-based therapy also avoids deleterious side effects (e.g., integration into chromosomes) that limit clinical application of most virus- and DNA-based vectors.² The first *in vivo* gene transfer therapy using mRNA was reported in 1990 (ref. 3). Subsequently, however, mRNA has rarely been used for introducing genetic material into animals or even into cultured cells. The use of mRNA has been mostly limited to vaccination in

which antigen-encoding transcripts were administered *in vivo* or delivered to dendritic cells (DCs) *ex vivo* in order to induce cellular and humoral immune responses.^{4,5} However, RNA is unsuitable for gene replacement because of its high immunogenicity and low effectiveness.

Recent studies have demonstrated that RNA activates cells of the innate immune system by stimulating Toll-like receptors (TLRs), specifically TLR3, TLR7, and TLR8.^{6–8} However, when naturally occurring modified nucleosides, for example, pseudouridine (Ψ), 5-methylcytidine (m5C), N6-methyladenosine (m6A), 5-methyluridine (m5U), or 2-thiouridine (s2U) were incorporated into the transcript, most of the TLRs were no longer activated.⁹ The immunostimulatory potential of such RNAs as measured by proinflammatory cytokine release and induction of co-stimulatory molecules was greatly diminished.⁹ Hartmann *et al.* also reported that modified nucleosides, including s2U and Ψ , abrogated 5'-triphosphate RNA-mediated activation of another RNA-responsive immune sensor, retinoic acid-inducible protein I (RIG-I).¹⁰ If any of the *in vitro* transcripts containing nucleoside modifications would remain translatable and also avoid immune activation *in vivo*, such an mRNA could be developed into a new therapeutic tool for both gene replacement and vaccination. Accordingly, in this report, we tested the nucleoside-modified mRNAs for their translational potentials and *in vivo* immune characteristics.

Our results reveal that incorporating pseudouridine, a naturally occurring modified nucleoside, into mRNA not only suppresses RNA-mediated immune activation *in vitro* and *in vivo*, but also enhances the translational capacity of the RNA. These characteristics and the ease of generating such an RNA by *in vitro* transcription make Ψ -containing mRNA a unique tool for expression of any protein *in vitro* and *in vivo*.

RESULTS

In vitro transcription and translation of nucleoside-modified mRNAs

In nature, RNA is synthesized from adenosine triphosphate, uridine triphosphate, cytidine triphosphate, and guanosine triphosphate. After transcription, selected nucleosides become modified.

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Due to technical limitations, at present, the simplest method to generate RNA with modified nucleosides is to perform *in vitro* transcription reactions in which one of the four basic nucleotide triphosphates is replaced with a corresponding modified nucleotide triphosphate.⁹ In these transcripts, one particular nucleotide is substituted with the modified nucleotide at every position. In a series of transcription reactions using reporter protein-encoding plasmids and RNA polymerases, we obtained full-length transcripts containing Ψ, m5U, s2U, m6A, or m5C (Figure 1a). Structures of these nucleosides are shown in Supplementary Figure S1. Under similar transcriptional conditions, we were unable to obtain full-length mRNA containing other naturally occurring nucleosides such as N1-methylguanosine, N1-methyladenosine, N7-methylguanosine, 2'-O-methyluridine, and 2'-O-methylcytidine.

First, a set of nucleoside-modified mRNAs encoding firefly luciferase (TEVlucA₅₀) and *Renilla* luciferase (capRen) (Supplementary Figure S2) was tested to determine whether they were translated in rabbit reticulocyte lysates. By measuring the enzyme activity of the encoded proteins, we were surprised to find that Ψ-containing mRNA translated more efficiently than its unmodified counterpart, resulting in twice as much protein from both TEVlucA₅₀ and capRen templates (Figure 1b). The translational yield of mRNAs with m5U or m5C was similar to the unmodified transcripts, while mRNAs containing s2U translated poorly. The presence of m6A in mRNAs completely abolished their translatability. The Ψ-mediated translational enhancement was not observed in all translational systems. In wheat germ extracts, ~50% less protein was produced from the Ψ-containing mRNAs compared with unmodified mRNAs, while in bacterial cell lysates, mRNAs with Ψ-modifications were not translated (Figure 1b). Analyses of radiolabeled translational products generated in reticulocyte lysates confirmed that lack of luciferase activities in samples programmed with s2U- and m6A-containing mRNAs were indeed due to lack of translation rather than loss of enzyme activity as a result of amino acid misincorporation caused by the modified nucleosides (Figure 1c). Although Ψ has not been found in any mRNA,¹¹ its ability to form hydrogen bonds with adenosine is similar to that of uridine (Figure 1d), a structural characteristic that should allow efficient translation of such an mRNA.

Superior translation of pseudouridine-containing mRNAs in cultured cells

To determine whether the nucleoside-modified mRNAs are translated in cultured cells, sets of *Renilla* luciferase-encoding mRNAs were lipofectin complexed and delivered into 293 cells. mRNA containing Ψ translated ~10 times as much as unmodified mRNA, while the presence of m5C led to a fourfold enhancement (Figure 2a). mRNAs with s2U or m6A constituents, as in the cell-free system, were poorly or not at all translated. The translational capacity of mRNA in which only 5% of the adenosines were substituted with m6A, a composition closer to that which occurs in nature,¹¹ was equivalent to that of the unmodified transcript. Similar results were obtained when sets of mRNAs with nucleoside modifications were tested in primary cells, including mouse DCs, embryonic fibroblasts and splenocytes, and human DCs (Figure 2b and data not shown). Control and Ψ-containing mRNAs delivered

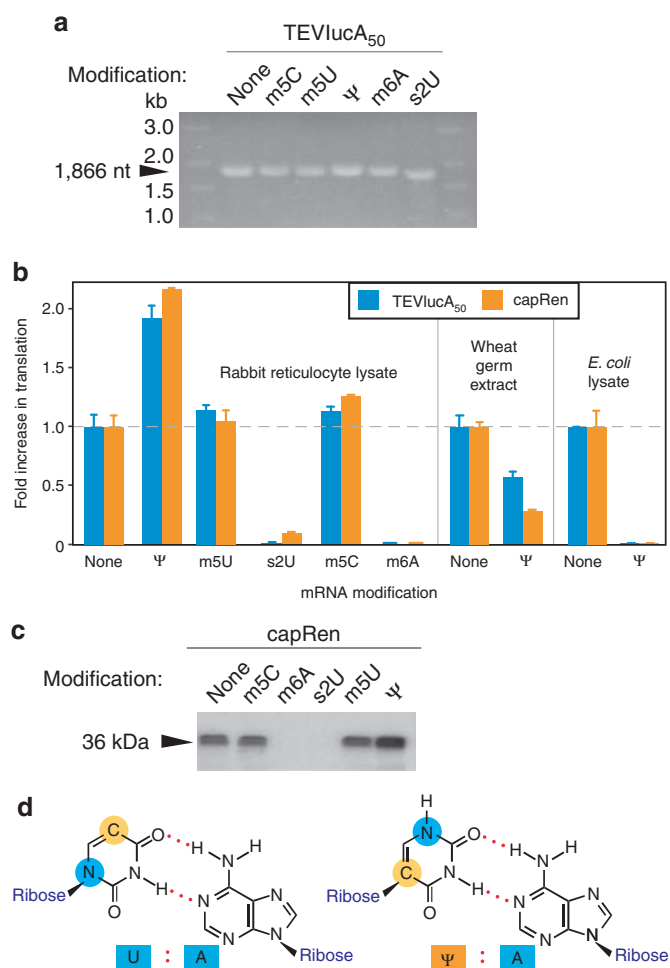


Figure 1 In vitro transcription and translation of nucleoside-modified mRNAs. (a) Aliquots of *in vitro*-transcribed TEVlucA₅₀ containing no or m5C-, m5U-, Ψ-, m6A-, or s2U-modified nucleosides were separated in a denaturing 1.4% agarose gel followed by ethidium bromide staining and ultraviolet illumination. (b) Rabbit reticulocyte lysate, wheat germ extract, and *Escherichia coli* S30 lysate were incubated in the presence of 50 ng/μl mRNA encoding firefly luciferase (TEVlucA₅₀) or *Renilla* luciferase (capRen). The mRNAs contained the indicated nucleoside modifications. Fold increase in translation was calculated by normalizing the measured relative light units to those obtained with nonmodified mRNA. Error bars indicate SEM ($n = 4$), and the dotted line represents the relative value obtained with unmodified mRNA in each of the lysates. (c) capRen mRNAs containing the indicated modification were translated in rabbit reticulocyte lysates supplemented with ³⁵S-labeled methionine. Aliquots of the lysates were separated by 15% sodium dodecyl sulfate-polyacrylamide gel electrophoresis and the fluorogram of the gel is shown. (d) Formation of hydrogen bonds (red dots) between uridine [U] and adenosine [A], or pseudouridine [Ψ] and adenosine [A]. Structural differences between uridine and pseudouridine are highlighted with blue- and yellow-filled circles.

by polyethyleneimine (PEI) gave results consistent with those obtained with lipofectin (Supplementary Figure S3).

In all of cell types, the Ψ modification conferred the highest translational yield to RNA and, because it was one of the nucleoside modifications that abolished RNA-mediated activation of primary DCs,⁹ as well as interferon (IFN) induction by RIG-I,¹⁰ subsequent studies focused on Ψ-modified mRNAs. To determine whether superior translation of Ψ-containing transcripts was dependent on

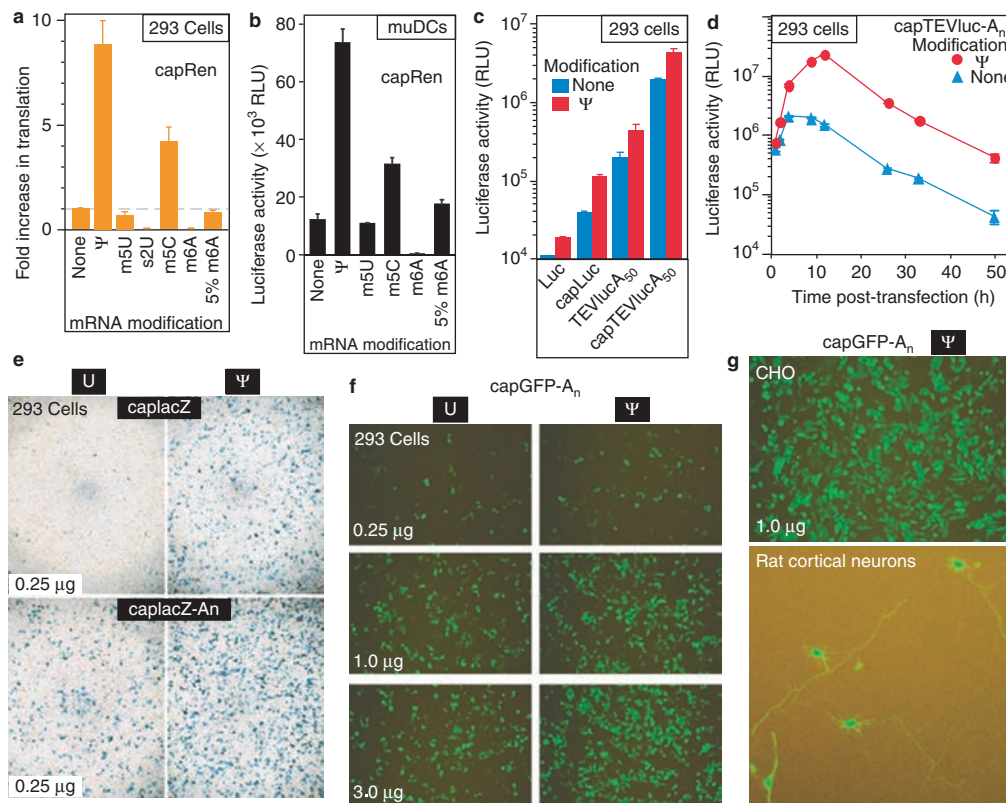


Figure 2 Translation of nucleoside-modified mRNAs in cells. (a–d) Cells seeded in 96-well plates were transfected with lipofectin-complexed mRNA (0.25 μ g/well), and levels of the encoded proteins were measured by enzymatic assays. (a) 293 cells (5×10^4 cells/well) and (b) bone marrow-derived murine dendritic cells (muDCs, 3×10^4 cells/well) were transfected with capped *Renilla* luciferase-encoding mRNA (capRen) containing the indicated nucleoside modification and then assayed at 4 and 8 hours, respectively. Values were either (a) normalized to the measured relative light units (RLU) obtained with nonmodified mRNA and expressed as fold increase, or (b) presented directly as RLU measured in one-tenth of the total 25 μ l lysate. (c) 293 cells (5×10^4 cells/well) were transfected with firefly luciferase-encoding mRNAs that had 5'-cap (capLuc), 50-nt long 3'-poly(A) tail (TEVlucA₅₀), both, or none of these elements (capTEVlucA₅₀ and Luc, respectively) and contained either Ψ or no nucleoside modifications. Cells were lysed 4 hours later and luciferase activities were measured in aliquots (1/20th) of the total 50 μ l lysates. (d) 293 cells (5×10^4 cells/well) transfected with capTEVlucA₅₀ containing unmodified or Ψ -modified nucleosides were lysed at the indicated time points following transfection. Aliquots (1/20th) of the 50 μ l lysates were assayed. Error bars in d are too small to visualize. (a–d) All values are expressed as the mean $\pm$ SEM ($n = 3$ –10/point). (a) Values are from three independently performed experiments or (b–d) one representative of at least three experiments. (e) 293 cells (5×10^4 cells/well) were transfected with lipofectin-complexed mRNAs (0.25 μ g/well) encoding bacterial β -galactosidase (lacZ). The transcripts had cap and 3'-poly(A) tail that were either 30-nt long (caplacZ) or ~200-nt long (caplacZ-A₅₀). Constructs containing unmodified uridine (U) or pseudouridine (Ψ) nucleosides were tested. Cells were fixed and stained with X-gal, 24 hours post-transfection. Images were taken by inverted microscopy (original magnification $\times 4$) from representative wells obtained in independently performed experiments ($n = 3$). (f) 293 cells were transfected with lipofectin-complexed capGFP-A₅₀ mRNAs (0.25–3.0 μ g/well, as indicated). The transcripts had cap and ~200-nt long poly(A) tail. Constructs containing unmodified uridine (U) or pseudouridine (Ψ) nucleosides were used. Cells were photographed at 24 hours post-transfection. Shown images are from representative wells obtained in one of the four independently performed experiments. (g) Expression of green fluorescent protein (GFP) in Chinese hamster ovary (CHO) cells and rat cortical neurons. CHO and rat neuronal cells, seeded at 5×10^4 cells/well density in 96- and 48-well plates, respectively, were transfected with lipofectin-complexed capGFP-A₅₀ containing pseudouridine (Ψ). Pictures taken at 24 hours post-transfection are representatives of 10 independently performed experiments.

structural elements known to enhance translation of the mRNA, a set of firefly luciferase-encoding mRNAs were synthesized with a unique 5'-untranslated sequence (TEV, a cap-independent translational enhancer¹²), cap and/or poly(A) tail, or no additional sequences. As expected, these structural elements enhanced translational efficiency of the mRNA and presence of Ψ in the transcripts further increased the amount of protein made from all tested constructs (Figure 2c), demonstrating that Ψ -mediated translational enhancement is independent of cap and poly(A) tail. The most efficiently translated mRNAs (capTEVlucA₅₀) were extended with longer poly(A) tails, which is known to enhance translation,¹² and analyzed in time-course studies. mRNA with Ψ modification was translated to a greater extent than unmodified RNA at every

tested time point and, as the incubation time increased, the difference between levels of luciferase made from the two types of mRNAs also increased (Figure 2d). Considering the short functional half-life of luciferase in mammalian cells and the fact that at 8–50 hours after transfection, cells programmed with Ψ -containing mRNA had unproportionally higher luciferase level, it is likely that Ψ -modification stabilizes the mRNA.

Routinely, β -galactosidase (lacZ)- and green fluorescent protein (GFP)-encoding constructs are used to visualize viral- or plasmid vector-mediated gene expression. Accordingly, we generated lacZ and GFP mRNAs (Supplementary Figure S2) with or without Ψ to use this detection system. To enhance translation, a subset of the RNAs was also extended with poly(A) tail. Analyses

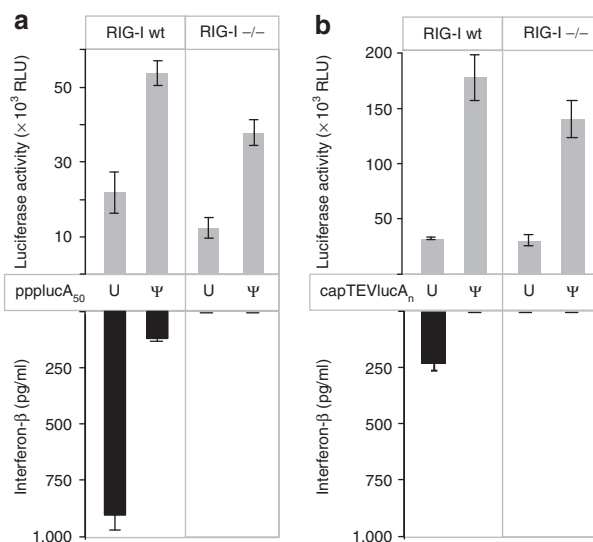


Figure 3 Superior translation of Ψ -modified mRNA is independent of retinoic acid-inducible protein 1 (RIG-I). Wild-type (wt) and RIG-I(-/-) mouse embryonic fibroblasts cultured in 96-well plate (5×10^4 cells/well) were transfected with lipofectin-complexed RNA (0.25 μ g/well). (a) The RNA (ppplucA₅₀) contained unmodified uridine (U) or pseudouridine (Ψ), and had triphosphate at its 5'-end. (b) The RNA (capTEVlucA₅₀) contained cap and long poly(A) tail. Upper panel, cells were lysed 4 hours later and luciferase activities were measured in the whole lysate. Values are measured in relative light units (RLU). Lower panel, cells were incubated for 24 hours and analyzed for interferon- β (IFN- β) by enzyme-linked immunosorbent assay. Values are expressed as the mean $\pm$ SEM ($n = 4$).

of 293 cells at 24 hours following RNA delivery demonstrated that the presence of Ψ in the mRNA significantly improved translational capacity of all tested RNA (caplacZ, caplacZ-A_n, and capGFP-A_n) as compared to the corresponding control, uridine-containing transcripts (Figure 2e and f). Not only did more cells express the encoded proteins when Ψ -modified mRNAs were used, but also the cells exhibited more intense staining, suggesting higher cellular levels of those proteins (Figure 2e and f and Supplementary Figure S4). The level of GFP staining in both number of positive cell and intensity was proportional to the amounts of delivered capGFP-A_n RNA added to cells cultured (Figure 2f). Transfection of GFP mRNA containing Ψ modification resulted in higher levels of GFP expression in Chinese hamster ovary (CHO) cells as well as in cultured primary neurons from rat brain cortex (Figure 2g).

Superior translation of Ψ -containing mRNA is independent of RIG-I

RIG-I, a cytoplasmic sensor of pathogenic RNA, is ubiquitously present in all cells.¹³ It directs type I IFN secretion when exposed to RNA containing 5'-triphosphate; however, replacement of uridine with Ψ in RNA blocks such an activation.¹⁰ The RNA samples we have synthesized and analyzed, including a significant fraction of those transcribed in the presence of cap analog, contained a 5'-triphosphate. This raises the possibility that the observed translational differences between U- and Ψ -containing RNA (Figure 2) were caused by type I IFN, secreted when RIG-I was exposed to uridine- but not to Ψ -containing RNA. To address the role of

RIG-I and type I IFN in the differential translation of the two RNA types, mouse embryonic fibroblasts (MEFs) derived from RIG-I(-/-) and wild-type mice were treated with U- or Ψ -containing mRNAs, and then translation and IFN induction were measured. Transcripts containing either 5'-triphosphates (ppplucA₅₀) or cap (capTEVlucA₅₀) were tested. Ψ -containing mRNA was translated at increased levels compared to U-containing mRNA independent from the presence of RIG-I (Figure 3). As expected, the level of IFN- β secreted by wild-type MEFs greatly diminished when the triphosphates were replaced with cap at the 5'-ends of U-containing RNAs (Figure 3a and b).

Most important, the U-containing mRNA did not induce IFN- β from the RIG-I(-/-) MEFs but was still translated less efficiently than Ψ -containing mRNA (Figure 3), making unlikely that translational differences were mediated by RIG-I-induced IFN actions.

Efficient translation of Ψ -containing mRNA in mice

For *in vivo* studies, we used firefly luciferase-encoding mRNA because, unlike other reporter enzymes (e.g., β -galactosidase, *Renilla* luciferase), no endogenous mammalian enzyme interferes with its detection. The transcripts, unmodified and Ψ -modified capTEVlucA₅₀, were extended with ~200-nt long poly(A) tails to increase their translatability (Figure 4a). mRNAs were lipofectin complexed and injected into mice through their caudal veins. Intravenous administration of 0.3- μ g capTEVluc-A_n containing Ψ resulted in high luciferase activity in the spleen, but not in the lung, liver, heart, kidney, or brain (Figure 4b), thus in subsequent studies only spleens were examined. Before analysis, spleens were bisected with one-half used for luciferase enzyme measurements and the other half for RNA isolation and subsequent northern blotting. First, the efficacies of Ψ -modified and unmodified capTEVluc-A_n mRNAs at 0.015 mg/kg (0.3 μ g/animal) dose were compared in time-course experiments. Luciferase activity was readily detectable at 1 hour, peaked at 4 hours, and declined by 24 hours after mRNA administration (Figure 4c). At 1 and 4 hours, significantly more (up to 12-fold) luciferase activity was detected in animals that received the Ψ -modified mRNA compared to those injected with the unmodified mRNA. By 24 hours, the only samples with detectable levels of luciferase (fourfold above background) were from animals injected with Ψ -containing mRNA. The overall expression patterns were similar when mRNAs encoding *Renilla* luciferase (capRen with or without Ψ modifications) were injected into the animals or when isolated mouse splenocytes were exposed to mRNAs in culture (Supplementary Figure S5). Northern analyses of splenic RNA revealed that the administered mRNAs, in their intact and partially degraded forms, were readily detectable at 15 minutes, 1 hour, and 4 hours postinjection (Figure 4d and Supplementary Figure S6). Because pilot studies demonstrated that the hybridization properties of Ψ -modified and unmodified RNAs are the same, we conclude that although equal amounts of the delivered mRNA reached the spleens, more Ψ -containing mRNA than control mRNA remained at 1 and 4 hours (Figure 4d and Supplementary Figure S6). By 24 hours, unmodified capTEVluc-A_n was below the level of detection, but trace amounts of Ψ -containing capTEVluc-A_n, though partially degraded, were detectable (Figure 4d). These results suggest that presence of Ψ in

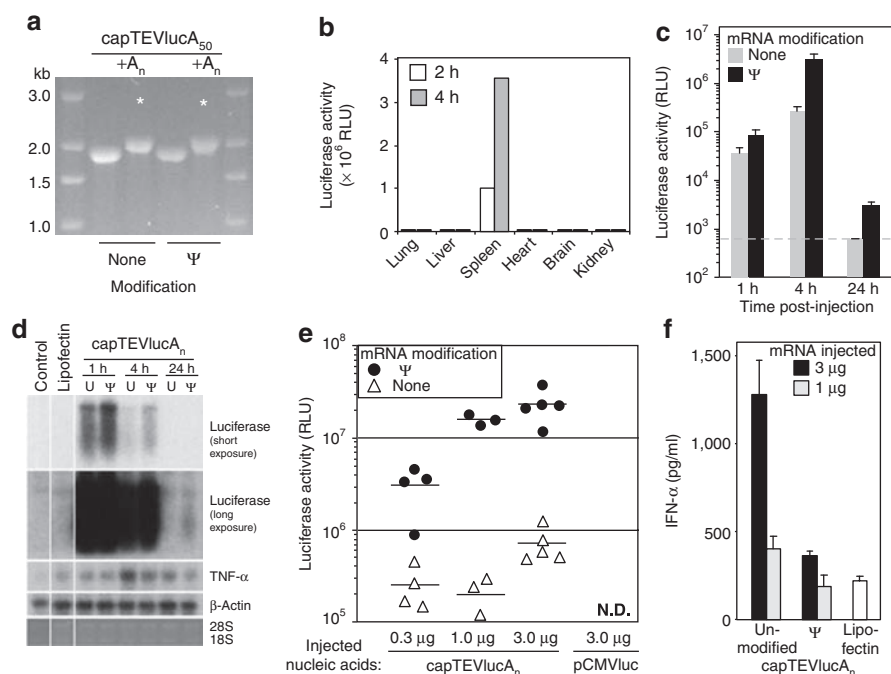


Figure 4 Ψ-modified mRNAs are nonimmunogenic and have a higher translational capacity than unmodified mRNA in mice. *In vitro*-transcribed capTEVlucA₅₀ (1,866 nt) with or without Ψ modifications were extended with long 3'-end poly(A) tail (+A_n) using poly(A) polymerase. Aliquots (1 μg) of mRNAs before and after poly(A) tailing were analyzed on denaturing agarose gel followed by ethidium bromide staining and ultraviolet (UV) illumination. The estimated lengths of poly(A) tails are ~200 nt. mRNAs used for the animal studies are indicated by asterisks. **(b)** Sixty-microliter aliquots of lipofectin-complexed mRNA (0.3 μg capTEVlucA_n/mouse) containing Ψ modifications were administered by caudal vein injection. Animals were killed at 2 and 4 hours postinjection and luciferase activities were measured in aliquots (1/10th) of organs homogenized in lysis buffer. Values represent luciferase activities in the whole organs. Results with capRen were quantitatively identical except that liver and kidney had high endogenous *Renilla* luciferase-like activities (data not shown). **(c,d)** Lipofectin-complexed capTEVlucA_n (0.3 μg/60 μl/animal) with or without Ψ modifications were intravenously (IV) delivered to mice. Animals were killed at 1, 4, and 24 hours postinjection and one-half of their spleens were processed for **(c)** luciferase measurements **(d)** while the other half for RNA analyses. Luciferase activities were measured in aliquots (1/5th) of the homogenate made from half of the spleens. Plotted values represent luciferase activities in the whole spleen and are expressed as the mean ± SEM. (*n* = 3 or 4/point). A similar pattern of expression was obtained in time-course experiments using capRen (data not shown). Dotted line represents background activity measured in spleen samples from animals injected only with lipofectin. **(d)** Aliquots of RNA (2 μg) isolated from the other half of spleens were analyzed by northern blot for luciferase, tumor necrosis factor-α (TNF-α) and β-actin. Autoradiograms and the corresponding ethidium bromide-stained, UV-visualized 28S and 18S rRNAs are presented. For luciferase, radiograms obtained after short and long exposure times (3 hours and 2 days) are shown. Animals, uninjected (control) and IV injected with uncomplexed lipofectin (lipofectin), were also processed. RNAs containing uridines (U) or pseudouridines (Ψ) are indicated. **(e)** The indicated amounts of lipofectin-complexed nucleic acids, capTEVlucA_n mRNA with or without Ψ constituents and pCMVluc plasmid DNA in a volume of 60 μl/animal were delivered by IV injection into mice. Animals injected with mRNA or plasmid DNA were killed at 6 or 24 hours postinjection, respectively, and luciferase activities were measured in aliquots (1/10th) of their spleens homogenized in lysis buffer. Presented values were adjusted to signify luciferase activities in the whole organs (*n* = 3–5/point). The value from each animal is shown and the short horizontal lines indicate the mean; ND, not detectable. **(f)** Serum samples, collected during killing (6 hours postinjection) from the same animals that were processed for luciferase assessment shown in **e**, were analyzed by enzyme-linked immunosorbent assay which revealed that 3 μg of unmodified mRNA induced a higher level of interferon-α (IFN-α) than 3 μg of Ψ-modified mRNA did (*P* < 0.001). The levels of IFN-α induced by 3 μg of Ψ-modified mRNA were similar to those obtained when animals were injected with uncomplexed lipofectin (*P* = 0.19). Values are expressed as the mean ± SEM (*n* = 3 or 5 animals/group).

RNA likely increases its biological stability. By altering the mRNA-lipid formulation, the delivery of greater amounts of mRNAs resulted in proportionally greater production of luciferase in cultured cells (**Supplementary Figure S7**). To determine whether protein production from the encoding mRNAs increases similarly *in vivo*, mRNAs were administered at 0.015–0.150 mg/kg doses, corresponding to delivery of 0.3–3.0 μg capTEVlucA_n per animal. Six hours later, the mice were killed and their spleens were analyzed. We found that luciferase levels were proportional to the amount of injected RNAs (**Figure 4e**). At these doses of RNA, luciferase levels were 12- to 78-fold higher when Ψ-modified capTEVlucA_n was injected compared to the corresponding unmodified capTEVlucA_n (**Figure 4e**). Again, no luciferase activity could be detected in any of the other organs even in animals injected with the highest

dose of mRNA. For comparison, 3.0 μg of pCMVluc plasmid was injected in a similar manner, but no luciferase activity could be detected in any organ when analyzed 24 hours postadministration (**Figure 4e** and data not shown).

Ψ-Modified mRNA is nonimmunogenic in mice

We recently demonstrated that unmodified RNA, in contrast to Ψ-modified RNA, activated human DCs to secrete IFN-α and tumor necrosis factor-α (TNF-α).⁹ In order to demonstrate whether Ψ-modified RNA activated the innate immune system after *in vivo* systemic delivery, serum samples were collected from animals at 6 hours postinjection and tested for proinflammatory cytokines. We found high IFN-α levels only in those animals that were injected with 3.0 μg of unmodified capTEVlucA_n, but not

with the Ψ -modified capTEVluc- A_n (Figure 4f). Although TNF- α protein was below the detection level in the collected serum samples, northern blot analysis demonstrated the highest amount of TNF- α mRNA in the spleen of animals that were administered unmodified mRNA at 4 hours postinjection (Figure 4d).

DISCUSSION

Progress in understanding the molecular details of many human diseases and instant access to human gene collections brings the possibility of using genes for treatment closer to reality. However, safe and effective gene therapy, which would take advantage of all these recent achievements, is still lacking. Unlike viral- and plasmid-based vectors, *in vitro*-transcribed mRNAs have never been vigorously tested for *in vivo* gene replacement, only for therapeutic vaccination (reviewed in ref. 14), because they were generally considered labile, inefficient, and immunogenic. Our most significant finding, however, demonstrates that *in vitro*-transcribed mRNAs containing pseudouridines possess superior translational capacities, increased biological stability, and no immunogenicity. The potential therapeutic advantages of using mRNA, especially Ψ -containing mRNA, over plasmid and viral vectors for delivering genetic material are numerous: (i) improved safety, because RNA is inherently incapable of integrating into the genome, thus preventing deleterious side effects that have stalled other vectors;² (ii) lack of inflammatory response to Ψ -containing mRNA, thereby avoiding devastating systemic inflammation that can be fatal with a viral vector;¹⁵ (iii) efficient transduction of primary cells, unlike DNA, RNA does not require cell proliferation for expression of the encoded protein; (iv) rapid protein expression, mRNAs are translated within minutes following entry into the cytoplasm, whereas plasmids require time-consuming nuclear import and transcription; (v) virtually no size limit for the encoded proteins, because long mRNAs (we have generated 12-kb long mRNAs) can be easily obtained, unlike viral vectors that possess limited packaging capacities; (vi) the extent and duration of the encoded protein expression can be closely controlled because mRNAs have shorter half lives and, unlike other vectors, do not replicate; and (vii) manufacturing of mRNA is much simpler than producing viral or plasmid vectors, one DNA template can be transcribed many (~100) times and, by immobilizing the template, the process is adaptable for large scale continuous production in bioreactors.¹⁶ Overall, these beneficial characteristics make Ψ -containing mRNA an excellent tool not only for *in vivo* expression of therapeutic proteins but also for vaccination. In the latter case, coadministration of adjuvant (e.g., immunostimulatory oligo-DNA, lipopolysaccharide) would also be required.

Our most surprising result is that mRNAs with Ψ modification have a higher translational capacity than those without modification in all tested mammalian systems. Although further studies are needed to understand the reason for this difference fully, using RIG-I(-/-) MEFs, we excluded a role for RIG-I and type I IFN in this phenomenon (Figure 3). It is possible that protein synthesis might be inhibited by RNA-dependent protein kinase activated by structural motifs present in mRNA-containing uridine¹⁷ but not Ψ modification. Consistent with this interpretation, attenuated translation of nonmodified mRNAs as compared to Ψ -modified mRNAs was observed in mammalian cells and lysates that contain

RNA-dependent protein kinase but not observed in wheat germ extracts (Figures 1b and 2), which have no RNA-dependent protein kinase.¹⁸

A likely contributing factor to the enhanced translation observed with Ψ modification is an increase in biological stability of the mRNAs (Figure 4d). Indeed, higher resistance to hydrolysis by phosphodiesterases from snake venom and spleen has been reported when uridine was replaced with Ψ in dinucleotide substrates.¹⁹ Previous studies have also demonstrated that Ψ stabilizes RNA secondary structures by promoting base stacking,²⁰ which could slow degradation. However, stability of mRNAs containing either uridines or pseudouridines was the same when tested by *in vitro* assays using human skin-associated RNases²¹ (data not shown). Enhanced translation might be another factor that improves stability by protecting the RNA with high ribosome occupancy.

From a potential adverse event standpoint, it is important to note that Ψ , a natural constituent of RNA, likely lacks toxicity. In fact, Ψ is the most common modification found in RNA and natural pathways for metabolism of Ψ -modified mRNA and subsequent disposal into the urine exist.²²

In previous studies, we demonstrated that bacterial RNA and *in vitro* transcripts without modified nucleoside constituents stimulated human TLR3, TLR7, and TLR8, while mammalian RNA and *in vitro* transcripts containing nucleoside modifications including Ψ were minimally or nonstimulatory.⁹ We also observed high levels of TNF- α and IFN- α secretion by human primary DCs when exposed to RNAs that lacked nucleoside modification, but not when Ψ -modified RNAs were used.⁹ Because mammalian RNA, unlike bacterial RNA, abundantly possesses modified nucleosides, these results collectively demonstrate that nucleoside modifications suppress RNA immunogenicity. This study extends these findings to *in vivo* conditions by showing induction of IFN- α secretion and an increase of TNF- α mRNA in mice following administration of unmodified, but not Ψ modified, mRNA (Figure 4d and f).

In summary, we demonstrate that mRNA containing Ψ is a new and effective vector to deliver genetic material for protein expression into both cultured cells and animals. The presence of Ψ in mRNA improved its translational capacity and overall stability. It also diminished its immunogenicity *in vivo*. Administering Ψ -containing mRNA at very low doses (0.015 mg/kg) resulted in robust expression of the encoded protein. We identified spleen as the target organ where the intravenously delivered, lipofectin-complexed mRNA and its encoded protein accumulate. These collective findings are important steps in developing the therapeutic potential of mRNA, such as using modified mRNA as an alternative to conventional vaccination and as a means for expressing clinically beneficial proteins *in vivo* safely and effectively.

MATERIALS AND METHODS

mRNA synthesis and characterization. For templates, reporter plasmids encoding firefly luciferase (pTEVluc, pTEVlucA₅₀, pLuc, and pLucA₅₀)¹² (Daniel Gallie, University of California at Riverside, CA), Renilla luciferase (pSVren),²³ bacterial β -galactosidase (lacZ) (Jay Hecker, University of Pennsylvania, PA), and enhanced GFP (peGFP) were used. Plasmid peGFP was generated by inserting coding sequences from pEGFP-N3 (Clontech, Mountain View, CA) into the EcoRV site of pT7TS (Paul Krieg, University of Texas at Austin, TX) containing the corresponding 5'- and 3'-untranslated

region sequences of *Xenopus* β -globin mRNA and a stretch of dA₃₀dC₃₀. First, plasmids were linearized with *Bam*HI (pTEVluc, pLuc), *Nde*I (pTEVlucA₅₀, pLucA₅₀), *Ssp*I (pSVren), or *Xba*I/*Sal*I (placZ, pGFP) to generate templates. Transcriptions were performed at 37°C for 3 hours using T7 or SP6 RNA polymerases and nucleotide triphosphates at 7.5 mmol/l final concentration (MEGAscript kits; Ambion, Austin, TX). To obtain mRNAs with modified nucleosides, the transcription reaction was assembled with the replacement of one nucleotide triphosphate with the corresponding triphosphate derivative of the following modified nucleosides: 5-methylcytidine (m5C), 5-methyluridine (m5U), 2-thiouridine (s2U), N6-methyladenosine (m6A), pseudouridine (Ψ), N1-methylguanosine (m1G), N1-methyladenosine (m1A), 2'-O-methyluridine (Um), 2'-O-methylcytidine (Cm) (TriLink, San Diego, CA), or N7-methylguanosine (m7G) (Sigma, St. Louis, MO). Where noted, only 5% of the adenosine triphosphate was replaced with N6-methyladenosine triphosphates in the reaction (5% m6A mRNA). Capped mRNAs were generated by supplementing the transcription reactions with 6 mmol/l 3'-O-Me-m7GpppG, a nonreversible cap analog, (New England Biolabs, Beverly, MA) and lowering the concentration of guanosine triphosphate (3.75 mmol/l). Selected mRNAs were also poly(A) tailed in a reaction of ~1.5 μ g/ μ l RNA, 5 mmol/l adenosine triphosphate, and 60 U/ μ l yeast poly(A) polymerase (USB, Cleveland, OH) according to the manufacturer's instructions and incubated at 30°C for 3 hours. The length of poly(A) tails were estimated to be ~200-nt long and is indicated with A_n. Purification of the transcripts were performed by Turbo DNase (Ambion, Austin, TX) digestion followed by LiCl precipitation and 75% ethanol wash. The concentrations of RNA reconstituted in water were determined by measuring the optical density at 260 nm. Efficient incorporation of modified nucleotides to the transcripts was demonstrated by HPLC analyses.⁹ All RNA samples were analyzed by denaturing agarose gel electrophoresis for quality assurance. Each RNA type was synthesized in 4–10 independently performed transcription experiments and all experiments were performed with at least two different batches of mRNA. Schematic representations of all mRNAs used in the study are shown in **Supplementary Figure S2**.

In vitro translation assay. Aliquots (9 μ l) of rabbit reticulocyte lysates, wheat germ extracts, or *Escherichia coli* S30 extracts (Promega, Madison, WI) were programmed with 1- μ l (0.5 μ g) TEVlucA₅₀ or capRen mRNAs and incubated for 60 minutes at 30, 25, or 37°C, respectively. Using cell culture lysis reagent (Promega, Madison, WI), the translation was stopped and aliquots were analyzed for enzyme activities using firefly and *Renilla* luciferase assay systems (Promega, Madison, WI) and a LUMAT LB 950 luminometer (Berthold/EG&G; Wallac, Gaithersburg, MD) at 10-second measuring time. Using recombinant firefly luciferase (Promega, Madison, WI), a standard curve was determined to be linear up to 3.6×10^7 relative light units measured activity, corresponding to 1.57-ng enzyme. All measurements were performed in the linear range of the standard curve. In selected experiments, the capRen-programmed rabbit reticulocyte lysates were also supplemented with ³⁵S-methionine (PerkinElmer, Waltham, MA). At the end of incubation, 1- μ l aliquots were removed to analyze the labeled *Renilla* luciferase by 15% polyacrylamide gel electrophoresis. The gel containing the labeled samples was treated with 1 mol/l Na salicylate, dried, and a fluorogram was generated by exposure to BioMax MS film (Kodak, Rochester, NY).

Cells. Human embryonic kidney 293 cells and dihydrofolate reductase(–/–) CHO cells were obtained from the American Type Culture Collection (Manassas, VA). Splenocytes were isolated from mouse (C57Bl/6) spleens using a cell strainer (70 μ m; BD Discovery Labware, Bedford, MA) and ACK lysis buffer (Sigma, St. Louis, MO). Embryonic fibroblasts from wild-type and RIG-I(–/–) mice,²⁴ 293 cells, and splenocytes were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with glutamine (Life Technologies, Gaithersburg, MD) and 10% fetal calf serum (HyClone, Logan, UT). CHO cells were cultured in Iscove's modified Dulbecco's medium supplemented with hypoxanthine (0.1 mmol/l),

thymidine (0.016 mmol/l), and 10% fetal calf serum. Murine DCs were generated from bone marrow cells of femurs and tibia of 6- to 9-week-old BALB/c mice as described.²⁵ The cells were cultured in RPMI (Life Technologies, Gaithersburg, MD) containing glutamine, 10% fetal calf serum, and 50 ng/ml muGM-CSF (R&D, Minneapolis, MN). Cells were maintained with fresh medium containing 50 ng/ml muGM-CSF every 2 days and used on day 7. Primary cortical culture was prepared from brain neocortices of E19 rat embryos as previously described,²⁶ cultured in DMEM supplemented with 10% fetal calf serum (HyClone), 10% Ham's F12 with glutamine (Whittaker Bioproducts, Walkersville, MD), and 50 U/ml penicillin–streptomycin (Sigma, St. Louis, MO).

Assembling the RNA complex. Lipofectin (Invitrogen, Carlsbad, CA) and mRNA were complexed in phosphate buffer that has been shown to enhance transfection *in vitro* and *in vivo*.^{27,28} To assemble a 50- μ l complex of RNA–lipofectin, first 0.4 μ l potassium phosphate buffer (0.4 mol/l, pH 6.2) containing 10 μ g/ μ l bovine serum albumin (Sigma, St. Louis, MO) was added to 6.7 μ l DMEM, then 0.8 μ l lipofectin was mixed in and the sample was incubated for 10 minutes. In a separate tube, 0.25–3.0 μ g of RNA was added to DMEM to a final volume of 3.3 μ l. Diluted RNA was added to the lipofectin mix and incubated for 10 minutes. Finally, the RNA–lipofectin complex was further diluted by adding 38.8 μ l DMEM. Fifty microliter of such a complex was used to transfect cells present in 1 well of a 96-well plate. RNA was complexed with PEI (jetPEI; Polyplus Transfection, Illkirch, France) according to the manufacturer's instructions. Cells were exposed to 50 μ l NaCl (150 mmol/l) containing 0.25 μ g RNA complexed with 0.5 μ l PEI.

Detection of reporter proteins in RNA-transfected cultured cells. A day before transfection, cells were seeded into 96-well plates. MEFs and 293 cells were seeded at 5.0×10^4 cells/well density, murine splenocytes and DCs at 4×10^5 and 3×10^4 cells/well, respectively. Unless stated otherwise, cells were exposed to 50 μ l DMEM containing lipofectin-complexed RNA (0.25 μ g) for 1 hour, which was then replaced with complete medium and further cultured. Cells were lysed in 25 μ l (primary cells) or 50 μ l (293 cells) firefly- or *Renilla*-specific lysis reagents (Promega). Aliquots of 2.5 or 10 μ l were assayed with the corresponding enzyme substrates described above. All of the 25- μ l cell lysates made from MEFs were used for the measurement. For visualizing β -galactosidase activity, 293 cells transfected with lipofectin-complexed caplacZ and caplacZ-A_n were processed at 24 hours after transfection using X-gal staining. Primary mixed rat cortical cells were seeded into 48-well poly-D-lysine-coated plate (Sigma, St. Louis, MO) at 5.0×10^4 cells/well density and transfected with 100 μ l lipofectin-complexed capGFP-A_n (1.0 μ g). Expression of GFP in 293, CHO, and neurons was documented using inverted epifluorescent Nikon microscope mounted with a Nikon D40 digital camera using a setting of ISO400 and 0.5-second exposure time.

Administering mRNA to mice. Animal studies were performed in accordance with the National Institutes of Health Guide for Care and Use of Laboratory Animals and approved by the Institutional Animal Care and Use Committee of the University of Pennsylvania. Female BALB/c mice (Charles River Laboratories, Wilmington, MA), 8-week-old, weighting 18–23 g were prewarmed using infrared light for 5 minutes and anesthetized with 3.5% halothane in a mixture of N₂O and O₂ (70%:30%) in an anesthetic chamber. Animals were then placed on a 37°C heating pad and maintained with 1% halothane in the mixture of N₂O and O₂ (70%:30%) using a nose mask until injection was completed. For each experimental condition, three to five animals were used. Complexes in 60- μ l final volume constructed with 1 μ l lipofectin and different amounts of nucleic acid (0.3, 1.0, or 3.0 μ g capTEVluc-A_n or 3.0 μ g pCMVluc²⁷) were injected into the lateral tail vein using a 1-ml syringe with a 27G½" needle (Becton Dickinson, San Diego, CA). Mice were killed when indicated (1, 4, 6, or 24 hours), and their spleens and blood were harvested. The spleens were

bisected horizontally with surgical scissors and processed immediately. Using a 1.5-ml Eppendorf tube with a plastic pestle (Kontes, Vineland, NJ), one-half was homogenized in 200 μ l of cell culture lysis reagents (Promega) and the other half in 1 ml of Trizol (Invitrogen, Carlsbad, CA). Tissue homogenates obtained with luciferase lysis buffers were spun for 30 seconds at 12,000g and luciferase activities were measured in 40- μ l aliquots of cleared supernatants. Tissue homogenates obtained with Trizol were used for RNA isolation according to the manufacturer's instructions. In selected experiments, to detect luciferase-specific activity and mRNA presence, other organs (liver, heart, lung, kidney, and brain) were also harvested and processed similar to the spleen.

Cytokine detection. Serum samples collected from mice 6 hours following lipofectin-complexed administration of 0.3, 1.0, or 3.0 μ g mRNA were diluted 1–10 in phosphate-buffered saline and analyzed for mouse IFN- α by enzyme-linked immunosorbent assay (PBL Biomedical Labs, Piscataway, NJ) and TNF- α (R&D, Minneapolis, MN). Culture medium collected from RIG-I(–/–) and wild-type MEFs at 24 hours after transfection with lipofectin-complexed RNA (0.25 μ g/well) were assayed without dilution for mouse IFN- β by enzyme-linked immunosorbent assay.

Northern blot analysis. Aliquots (2.0 μ g) of RNA samples isolated from spleen homogenized in Trizol were separated by denaturing, 1.4% agarose gel electrophoresis, transferred to charged membranes (Schleicher and Schuell, Keene, NH) and hybridized in MiracleHyb (Stratagene, La Jolla, CA) as previously described.²³ Plasmids containing firefly luciferase (pLuc), rat-specific β -actin (accession no. AA900159; Open Biosystems, Huntsville, AL), and rat-specific TNF- α (pTOPO-TNF) cDNAs were used to gel-purify inserts for probes. pTOPO-TNF was generated by TOPO-TA cloning (Invitrogen, Carlsbad, CA) using rat brain RNA and TNF- α -specific PCR product, a 5'-primer (5'-CAGAACTCCAGGCGGTGTC-3') and 3'-primer (5'-AGTAGACCTGCCCGACTC-3') corresponding to nt 73–91 and nt 688–670 of the coding sequence of rat TNF- α (accession: NM_012675). The specificity of all probes was confirmed by sequencing. To probe the membranes, 50 ng of DNA was labeled using Redivue [32 P] dCTP (Amersham, Piscataway, NJ) with a random prime labeling kit (Boehringer Mannheim, Indianapolis, IN). Hybridized membranes were exposed to Kodak BioMax MS film using an MS intensifier screen at –70°C.

Statistical analysis. All data are reported as mean $\pm$ SEM. Statistical differences between treatment groups were calculated using Microsoft Excel. For all statistical testing, a *P* value <0.05 was considered significant.

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SUPPLEMENTARY MATERIAL

Figure S1. Structures of unmodified and modified nucleosides used in the study.

Figure S2. Schematic representation of mRNAs used in this study.

Figure S3. Expression of *Renilla* luciferase in 293 cells transfected with lipofectin- or PEI-complexed capRen.

Figure S4. Expression of β -galactosidase in 293 cells transfected with the encoding mRNAs analyzed at high power.

Figure S5. Translation of mRNAs transfected into mouse splenocytes.

Figure S6. Ψ -modified and unmodified mRNAs reach the spleen with the same efficiency.

Figure S7. Dose-dependent expression of luciferase in 293 cells transfected with mRNA.

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